14

Wave–particle nonlinearities

14.1 Introduction

As seen in Chapters 4 through 7, linear models of plasma wave dynamics are straightforward and rich in descriptive power. The principle of superposition constitutes the very essence of linearity and underlies all concepts and methods inherent to linear models. In particular, the concepts of eigenmodes, eigenvalues, eigenvectors, orthogonality, and the method of integral transforms all ultimately depend on the validity of the principle of superposition.

While many important phenomena are adequately characterized by linear models, there nevertheless exist many other important phenomena where this is not so because the principle of superposition breaks down either partially or completely. Typically the phenomenon in question becomes amplitude-dependent above some critical amplitude threshold and then nonlinearity becomes important. Breakdown of the superposition principle means that modes with different eigenvalues (to the extent eigenvalues still exist) are no longer independent and start to interact with each other. These nonlinear interactions result when products of dependent variables become important in the system of equations. As an example, consider the product of two modes having respective frequencies ω_1 and ω_2. This product can be decomposed into sum and difference frequencies according to standard trigonometric identities, e.g.,

$$\cos(\omega_1 t)\cos(\omega_2 t) = \frac{\cos\left[(\omega_1+\omega_2)t\right]+\cos\left[(\omega_1-\omega_2)t\right]}{2}. \tag{14.1}$$

The beat waves at frequencies $\omega_1 \pm \omega_2$ can act as source terms driving oscillations at $\omega_1 \pm \omega_2$. In the special case where $\omega_1 = \omega_2$, the difference frequency is zero and the nonlinear product can act as a source term that modulates equilibrium parameters and thereby changes the mode dynamics. In particular, feedback loops can develop where modes affect their own stability properties. Another possibility occurs when the nonlinear product is very small, but happens to resonantly drive

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a linear mode. In this case even a weak nonlinear coupling between two modes can resonantly drive another linear mode to large amplitude. Similar beating can occur with spatial factors $\sim \exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x})$ so that the nonlinear product of modes with wavevectors $\mathbf{k}_1$ and $\mathbf{k}_2$ drives spatial oscillations with wavevectors $\mathbf{k}_1 \pm \mathbf{k}_2$. In analogy to quantum mechanics the momentum associated with a wave is found to be proportional to $\mathbf{k}$ and the energy proportional to ω. The beating together of two waves with respective space-time dependence $\exp(\mathrm{i}\mathbf{k}_1\cdot\mathbf{x} - \mathrm{i}\omega_1 t)$ and $\exp(\mathrm{i}\mathbf{k}_2\cdot\mathbf{x} - \mathrm{i}\omega_2 t)$ can then be interpreted in terms of conservation of wave energy and momentum,

$$\begin{aligned}\omega_3 &= \omega_1 + \omega_2 \\ \mathbf{k}_3 &= \mathbf{k}_1 + \mathbf{k}_2,\end{aligned} \tag{14.2}$$

where wave 3 is what results from beating together waves 1 and 2.

Because of the large variety of possible nonlinear effects, there are many ways to categorize nonlinear behavior. For example, one categorization is according to whether the nonlinearity involves velocity space (Vlasov nonlinearity) or position space (fluid nonlinearity). Vlasov nonlinearities are characterized by energy exchange between wave electric fields, resonant particles, and non-resonant particles. Fluid instabilities involve nonlinear mixing of two or more fluid modes and can be interpreted as one wave modulating the equilibrium seen by another wave. Another categorization is according to whether the nonlinearity is weak or strong. In situations where the nonlinear coupling is weak, linear theory may be invoked as a reasonable first approximation and then used as a basis for developing the nonlinear model. In situations where the nonlinear coupling is strong, linear assumptions fail completely and the nonlinear behavior must be addressed directly without any help from linear theory. Weak nonlinear theories can be further categorized into (i) mode-coupling models, where a small number of modes mutually interact in a coherent manner and (ii) weak turbulence models, where a statistically large number of modes mutually interact with random phase so some sort of averaging is required.

An important feature of nonlinear theory concerns energy. Because energy is a quadratic function of amplitude, energy does not satisfy the principle of superposition and so energy cannot be properly accounted for in linear models. Thus, nonlinear models are essential for tracking the flow of energy between modes and also between modes and the equilibrium.

This chapter will consider nonlinearities involving velocity space so a Vlasov description is required and wave–particle interactions are important. The following chapter will consider wave–wave nonlinearities and so will involve a fluid description of the plasma.

14.2 Vlasov nonlinearity and quasi-linear velocity space diffusion

14.2.1 Derivation of the quasi-linear diffusion equation

Quasi-linear theory (Vedenov, Velikhov, and Sagdeev 1962, Drummond and Pines 1962, Bernstein and Engelmann 1966), a surprisingly complete extension to the Landau model of plasma waves, shows how plasma waves can alter the equilibrium velocity distribution. In order to focus attention on the most essential features of this theory, a simplified situation will be considered where the plasma is assumed to be one dimensional, uniform, and unmagnetized. Furthermore, only electrostatic modes will be considered and, by assuming the ions are infinitely massive, ion motion will be neglected. Thus, the plasma is characterized by the coupled Vlasov and Poisson equations for electrons and the ions simply provide a static, uniform neutralizing background. It is assumed that the electron velocity distribution function can be decomposed into (i) a spatially independent equilibrium term, which is allowed to have a slow temporal variation, and (ii) a small, high-frequency perturbation having a space-time dependence resulting from a spectrum of linear plasma waves of the sort discussed in Section 5.3. Thus, it is assumed that the electron velocity distribution has the form

$$f(x, v, t) = f_0(v, t) + f_1(x, v, t) + f_2(x, v, t) + \ldots, \tag{14.3}$$

where it is implicit that the magnitude of terms with subscript n is of order ϵ^n, where $\epsilon \ll 1$. At $t = 0$, the terms f_n, where $n \geq 2$, all vanish because the perturbation was prescribed to be f_1 at $t = 0$. Other variables such as the electric field will have some kind of nonlinear dependence on the distribution function and so, for example, the electric field will have the form

$$E = E_0 + E_1 + E_2 + E_3 + \ldots, \tag{14.4}$$

where it is again implicit that the nth term is of order ϵ^n.

Since by assumption, $f_0(v, t)$ does not depend on position, it is convenient to define a velocity-normalized order zero distribution function

$$f_0(v, t) = n_0 \bar{f}_0(v, t) \tag{14.5}$$

so

$$\int \bar{f}_0(v, t) \mathrm{d}v = 1. \tag{14.6}$$

This definition causes n_0 to show up explicitly so terms such as $\left(e^2/m\varepsilon_0\right) \partial f_0/\partial v$ can be written as $\omega_p^2 \partial \bar{f}_0/\partial v$. A possible initial condition for $f_0(v, t)$, namely a monotonically decreasing velocity distribution, is shown in Fig. 14.1(a).

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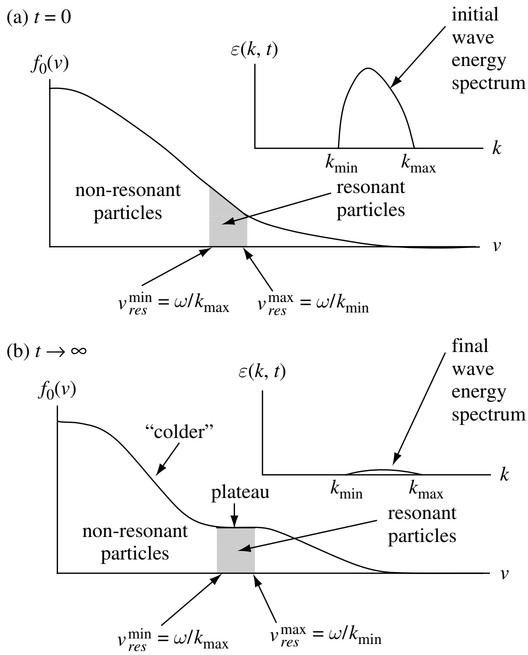

Fig. 14.1 (a) At $t = 0$ the equilibrium distribution function $f_0(v)$ is monotonically decreasing resulting in Landau damping of any waves and there is a wave spectrum (insert) with wave energy in the spectral range $k_{\min} < k < k_{\max}$. Resonant particles are shown as shaded in distribution function and lie in velocity range $v_{res}^{\min} < v < v_{res}^{\max}$. (b) As $t \to \infty$ the resonant particles develop a plateau (corresponding to absorbing energy from the wave), the wave spectrum goes to zero, and the non-resonant particles appear to become colder.

Spatial averaging will be used in the mathematical procedure to filter out some types of terms while retaining others. This averaging will be denoted by $\langle\rangle$ so, for example, the average of f_1 is

$$\langle f_1(x, v, t)\rangle = \frac{1}{L}\int \mathrm{d}x\, f_1(x, t), \tag{14.7}$$

where L is the length of the one-dimensional system and the integration is over this length. The spatial average of a quantity is independent of x and so, since f_0 is assumed to be independent of position, $\langle f_0\rangle = f_0$. (In an alternate version of the theory, the averaging is instead understood to be over a statistically large ensemble of systems containing turbulent waves and, in this version, the averaged quantity can be position-dependent.)

The 1-D electron Vlasov equation is

$$\frac{\partial}{\partial t}(f_0+f_1+f_2+\ldots)+v\frac{\partial}{\partial x}(f_1+f_2+\ldots)$$
$$-\frac{e}{m}(E_1+E_2+\ldots)\frac{\partial}{\partial v}(f_0+f_1+f_2+\ldots)=0; \tag{14.8}$$

there is no $\partial f_0/\partial x$ term because f_0 is assumed to be spatially independent and also there is no E_0 term because the system is assumed to be neutral in equilibrium. The linear portion of this equation,

$$\frac{\partial f_1}{\partial t}+v\frac{\partial f_1}{\partial x}-\frac{e}{m}E_1\frac{\partial f_0}{\partial v}=0, \tag{14.9}$$

forms the basis for the linear Landau theory of plasma waves discussed in Section 4.5. Subtracting Eq. (14.9) from Eq. (14.8) leaves the remainder equation

$$\frac{\partial}{\partial t}(f_0+f_2+\ldots)+v\frac{\partial}{\partial x}(f_2+\ldots)-\frac{e}{m}(E_2+\ldots)\frac{\partial}{\partial v}(f_0+f_1+f_2+\ldots)$$
$$-\frac{e}{m}E_1\frac{\partial}{\partial v}(f_1+f_2+\ldots)=0. \tag{14.10}$$

We assume quantities with subscripts $n\geq 1$ are waves and therefore have spatial averages that vanish, i.e., $\langle f_1\rangle=0$, $\langle E_1\rangle=0$, etc. Also, since f_0 is independent of position, $\langle E_2 f_0\rangle=f_0\langle E_2\rangle=0$, etc. Spatial averaging of Eq. (14.10) thus annihilates many terms, leaving

$$\frac{\partial f_0}{\partial t}-\frac{e}{m}\frac{\partial}{\partial v}\left[\langle E_1 f_1\rangle+\langle E_2 f_1\rangle+\ldots\right], \tag{14.11}$$

where $\partial/\partial v$ has been factored out of the spatial averaging because v is an independent variable in phase-space. The term $\langle E_1 f_1\rangle$ is of order ϵ^2 whereas $\langle E_2 f_1\rangle$ is of order ϵ^3 and the terms represented by the $+\ldots$ are of still higher order. The essential postulate of quasi-linear theory is that all terms of order ϵ^3 and higher may be neglected because ϵ is small. On applying this postulate, Eq. (14.11) reduces to the *quasi-linear velocity-space diffusion equation*

$$\frac{\partial f_0}{\partial t}=\frac{e}{m}\frac{\partial}{\partial v}\langle E_1 f_1\rangle. \tag{14.12}$$

This implies that even though f_0 is of order ϵ^0, the time derivative of f_0 is of order ϵ^2 so f_0 is a very slowly changing equilibrium.

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The nonlinear term $\langle E_1 f_1\rangle$ can be explicitly calculated by first expressing the perturbations as a sum (i.e., integral) over spatial Fourier modes, for example the first-order electric field can be expressed as

$$E_1(x,t) = \frac{1}{2\pi}\int \mathrm{d}k\,\tilde{E}_1(k,t)\mathrm{e}^{\mathrm{i}kx}. \tag{14.13}$$

This allows the product on the right-hand side of Eq. (14.12) to be evaluated as

$$\begin{aligned}\langle E_1 f_1\rangle &= \frac{1}{L}\int \mathrm{d}x\left[\left(\frac{1}{2\pi}\int \mathrm{d}k\,\tilde{E}_1(k,t)\mathrm{e}^{\mathrm{i}kx}\right)\left(\frac{1}{2\pi}\int \mathrm{d}k'\,\tilde{f}_1(k',v,t)\mathrm{e}^{\mathrm{i}k'x}\right)\right]\\ &= \frac{1}{2\pi L}\int \mathrm{d}k\int \mathrm{d}k'\tilde{E}_1(k,t)\,\tilde{f}_1(k',v,t)\frac{1}{2\pi}\int \mathrm{d}x\mathrm{e}^{\mathrm{i}(k+k')x},\end{aligned} \tag{14.14}$$

where the order of integration has been changed in the second line. Then, invoking the representation of the Dirac delta function

$$\delta(k) = \frac{1}{2\pi}\int \mathrm{d}x\,\mathrm{e}^{\mathrm{i}kx}, \tag{14.15}$$

Eq. (14.14) reduces to

$$\langle E_1 f_1\rangle = \frac{1}{2\pi L}\int \mathrm{d}k\,\tilde{E}_1(-k,t)\,\tilde{f}_1(k,v,t). \tag{14.16}$$

The linear perturbations E_1 and f_1 are governed by the system of linear equations discussed in Section 4.5. This means that associated with each wavevector k there is a complex frequency $\omega(k)$, which is determined by the linear dispersion relation $\mathcal{D}(\omega(k),k)=0$. This gives the explicit time dependence of the modes,

$$\tilde{E}_1(k,t) = \tilde{E}_1(k)\mathrm{e}^{-\mathrm{i}\omega(k)t} \tag{14.17a}$$

$$\tilde{f}_1(k,v,t) = \tilde{f}_1(k,v)\mathrm{e}^{-\mathrm{i}\omega(k)t}. \tag{14.17b}$$

Furthermore, Eq. (14.9) provides a relationship between $\tilde{E}_1(k)$ and $\tilde{f}_1(k,v)$ since the spatial Fourier transform of this equation is

$$-\mathrm{i}\omega f_1 + \mathrm{i}kvf_1 - \frac{e}{m}E_1\frac{\partial f_0}{\partial v} = 0, \tag{14.18}$$

which leads to the familiar linear relationship

$$\tilde{f}_1(k,v) = \mathrm{i}\frac{e}{m}\tilde{E}_1(k)\frac{\partial f_0/\partial v}{\omega - kv}. \tag{14.19}$$

Inserting Eqs. (14.19) and (14.17a) into Eq. (14.16) gives

$$\langle E_1 f_1\rangle = \frac{\mathrm{i}}{2\pi L}\frac{e}{m}\int \mathrm{d}k\,\tilde{E}_1(-k)\tilde{E}_1(k)\frac{\partial f_0/\partial v}{\omega - kv}\mathrm{e}^{-\mathrm{i}[\omega(-k)+\omega(k)]t}. \tag{14.20}$$

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This expression can be further evaluated by invoking the parity properties of $\tilde{E}_1(k)$ and $\omega(k)$. These parity properties are established by writing

$$E_1(x,t) = \frac{1}{2\pi}\int \mathrm{d}k\, \tilde{E}_1(k) \mathrm{e}^{\mathrm{i}kx - \mathrm{i}\omega(k)t} \tag{14.21}$$

and noting that the left-hand side is real because it describes a physical quantity. Taking the complex conjugate of Eq. (14.21) gives

$$E_1(x,t) = \frac{1}{2\pi}\int \mathrm{d}k\, \tilde{E}_1^*(k) \mathrm{e}^{-\mathrm{i}kx + \mathrm{i}\omega^*(k)t}. \tag{14.22}$$

The parity properties are determined by defining a temporary new integration variable $k' = -k$ and noting that $\int_{-\infty}^{\infty} \mathrm{d}k$ corresponds to $\int_{-\infty}^{\infty} \mathrm{d}k'$ since $\mathrm{d}k' = -\mathrm{d}k$ and the k limits $(-\infty, \infty)$ correspond to the k' limits $(\infty, -\infty)$. Thus Eq. (14.22) can be recast as

$$\begin{aligned} E_1(x,t) &= \frac{1}{2\pi}\int \mathrm{d}k'\, \tilde{E}_1^*(-k') \mathrm{e}^{\mathrm{i}k'x + \mathrm{i}\omega^*(-k')t} \\ &= \frac{1}{2\pi}\int \mathrm{d}k\, \tilde{E}_1^*(-k) \mathrm{e}^{\mathrm{i}kx + \mathrm{i}\omega^*(-k)t}, \end{aligned} \tag{14.23}$$

where the primes have now been removed in the second line. Since Eq. (14.21) and (14.23) have the same left-hand sides, the right-hand sides must also be the same. Because $\tilde{E}_1(k)$ and $\omega(k)$ are arbitrary, they must individually satisfy the respective parity conditions

$$\begin{aligned} \tilde{E}_1(k) &= \tilde{E}_1^*(-k) \\ \omega(k) &= -\omega^*(-k) \end{aligned} \tag{14.24}$$

or equivalently

$$\begin{aligned} \tilde{E}_1(-k) &= \tilde{E}_1^*(k) \\ \omega(-k) &= -\omega^*(k). \end{aligned} \tag{14.25}$$

If the complex frequency is written in terms of explicit real and imaginary parts $\omega(k) = \omega_r(k) + \mathrm{i}\omega_i(k)$ then the frequency parity condition becomes

$$\begin{aligned} \omega_r(-k) + \mathrm{i}\omega_i(-k) &= -\left[\omega_r(k) + \mathrm{i}\omega_i(k)\right]^* \\ &= -\omega_r(k) + \mathrm{i}\omega_i(k) \end{aligned} \tag{14.26}$$

from which it can be concluded

$$\begin{aligned} \omega_r(-k) &= -\omega_r(k) \\ \omega_i(-k) &= \omega_i(k). \end{aligned} \tag{14.27}$$

The real part of ω is therefore an odd function of k whereas the imaginary part is an even function of k.

Application of the parity conditions gives

$$\tilde{E}_1(-k)\tilde{E}_1(k) = \tilde{E}_1^*(k)\tilde{E}_1(k) = \left|\tilde{E}_1(k)\right|^2 \tag{14.28}$$

and

$$\omega(-k) + \omega(k) = 2\mathrm{i}\omega_i(k), \tag{14.29}$$

in which case Eq. (14.20) reduces to

$$\langle E_1 f_1 \rangle = \frac{\mathrm{i}}{2\pi L}\frac{e}{m}\int \mathrm{d}k \, \frac{\left|\tilde{E}_1(k)\right|^2 \mathrm{e}^{2\omega_i(k)t}}{\omega - kv}\frac{\partial f_0}{\partial v}. \tag{14.30}$$

Substitution of $\langle E_1 f_1 \rangle$ into Eq. (14.11) gives the time evolution of the equilibrium distribution function,

$$\frac{\partial f_0}{\partial t} = \frac{\mathrm{i}}{2\pi L}\frac{e^2}{m^2}\frac{\partial}{\partial v}\int \mathrm{d}k \, \frac{\left|\tilde{E}_1(k)\right|^2 \mathrm{e}^{2\omega_i(k)t}}{\omega - kv}\frac{\partial f_0}{\partial v}. \tag{14.31}$$

The physical meaning of the quantity $\left|\tilde{E}_1(k)\right|^2$ can be understood by considering the volume average of the electric field energy

$$\begin{aligned}
\langle W_E \rangle &= \left\langle \frac{\varepsilon_0 E_1^2}{2} \right\rangle \\
&= \frac{\varepsilon_0}{2L}\int \mathrm{d}x \left(\frac{1}{2\pi}\int \mathrm{d}k \, \tilde{E}_1(k)\mathrm{e}^{\mathrm{i}kx - \mathrm{i}\omega(k)t}\right)\left(\frac{1}{2\pi}\int \mathrm{d}k' \, \tilde{E}_1(k')\mathrm{e}^{\mathrm{i}k'x - \mathrm{i}\omega(k')t}\right) \\
&= \frac{\varepsilon_0}{4\pi L}\int \mathrm{d}k \int \mathrm{d}k' \, \tilde{E}_1(k)\tilde{E}_1(k')\mathrm{e}^{-\mathrm{i}[\omega(k)+\omega(k')]t}\frac{1}{2\pi}\int \mathrm{d}x \, \mathrm{e}^{\mathrm{i}(k+k')x}.
\end{aligned} \tag{14.32}$$

However, using Eq. (14.15) this can be written as

$$\begin{aligned}
\langle W_E \rangle &= \frac{\varepsilon_0}{4\pi L}\int \mathrm{d}k \, \tilde{E}_1(k)\tilde{E}_1(-k)\mathrm{e}^{-\mathrm{i}[\omega(k)+\omega(-k)]t} \\
&= \int \mathrm{d}k \, \mathcal{E}(k,t),
\end{aligned} \tag{14.33}$$

where

$$\mathcal{E}(k,t) = \frac{\varepsilon_0}{4\pi L}\left|\tilde{E}_1(k)\right|^2 \mathrm{e}^{2\omega_i(k)t} \tag{14.34}$$

is the time-dependent electric field energy density associated with the wavevector k. A possible initial condition for the distribution function and wave-energy spectrum is shown in Fig. 14.1(a); the wave-energy spectrum is shown in the insert and is finite in the range $k_{\min} < k < k_{\max}$. The distribution function is monotonically decreasing so as to cause Landau damping of the waves.

Combination of Eqs. (14.31) and (14.34) shows the evolution of the equilibrium distribution function is given by

$$\frac{\partial f_0}{\partial t} = \frac{2\mathrm{i}}{\varepsilon_0}\frac{e^2}{m^2}\frac{\partial}{\partial v}\int \mathrm{d}k\,\frac{\mathcal{E}(k,t)}{\omega - kv}\frac{\partial f_0}{\partial v}. \tag{14.35}$$

This can be summarized as a velocity-space diffusion equation

$$\frac{\partial f_0}{\partial t} = \frac{\partial}{\partial v}\left(D_{QL}\frac{\partial f_0}{\partial v}\right), \tag{14.36}$$

where the *quasi-linear velocity-space diffusion coefficient* is

$$D_{QL} = \frac{2\mathrm{i}e^2}{\varepsilon_0 m^2}\int \mathrm{d}k\,\frac{\mathcal{E}(k,t)}{\omega - kv}. \tag{14.37}$$

The factor i in the velocity diffusion tensor seems surprising because f_0 is a real quantity. However, the i factor is entirely appropriate and is intimately related to the parity properties of $\omega(k)$ since

$$\frac{\mathrm{i}}{\omega(k) - kv} = \frac{\mathrm{i}\left[\omega_r(k) - kv\right] + \omega_i(k)}{\left[\omega_r(k) - kv\right]^2 + \omega_i^2(k)}. \tag{14.38}$$

The denominator in this expression is an even function of k, the term $[\omega_r(k) - kv]$ in the numerator is an odd function of k, and $\omega_i(k)$ is an even function of k. Since $\mathcal{E}(k,t)$ is an even function of k, integration over k in Eq. (14.37) annihilates the imaginary component (this component is an odd function of k) and so

$$D_{QL} = \frac{e^2}{\varepsilon_0 m^2}\int \mathrm{d}k\,\frac{2\omega_i(k)\mathcal{E}(k,t)}{\left[\omega_r(k) - kv\right]^2 + \omega_i^2(k)}. \tag{14.39}$$

In summary, the self-consistent coupled system of equations for the nonlinear evolution of the equilibrium consists of Eq. (14.36), (14.39), and

$$\frac{\partial}{\partial t}\mathcal{E}(k,t) = 2\omega_i(k)\mathcal{E}(k,t), \tag{14.40}$$

which is obtained from Eq. (14.34). The real and imaginary parts of the frequency $\omega_r(k)$, $\omega_i(k)$ appear as parameters in these equations and are determined from the linear wave dispersion relation, which in turn depends on f_0. This linear wave dispersion relation is obtained as in Section 4.5 by writing $E_1 = -\partial\phi_1/\partial x$ and then combining Eq. (14.19) with Poisson's equation

$$k^2\tilde{\phi}_1(k) = -\frac{e}{\varepsilon_0}\int \mathrm{d}v\tilde{f}_1 \tag{14.41}$$

to obtain

$$k^2\tilde{\phi}_1(k) = -\frac{e^2}{\varepsilon_0 m}\int \mathrm{d}v\frac{\partial f_0}{\partial v}\frac{k\tilde{\phi}_1(k)}{\omega - kv}, \tag{14.42}$$

which can be expressed, using Eq. (14.5), as

$$1+\frac{\omega_p^2}{k}\int \mathrm{d}v\frac{\partial \bar{f}_0/\partial v}{\omega-kv}=0. \tag{14.43}$$

Thus, given the instantaneous value of $f_0(v,t)=n_0\bar{f}_0(v,t)$, the complex frequency is determined from Eq. (14.43) and then, given the instantaneous value of the wave spectral energy, both the evolution of f_0 and the wave spectral energy are determined from Eqs. (14.36) and (14.40).

14.2.2 Conservation properties of the quasi-linear diffusion equation

Conservation of particles

Conservation of particles occurs automatically because Eq. (14.36) has the form of a derivative in velocity space. Thus, the zeroth moment of Eq. (14.36) is simply

$$\frac{\partial}{\partial t}\int \mathrm{d}vf_0=\int \mathrm{d}v\frac{\partial}{\partial v}\left(D_{QL}\frac{\partial f_0}{\partial v}\right)=\left[D_{QL}\frac{\partial f_0}{\partial v}\right]_{v=-\infty}^{v=\infty}=0 \tag{14.44}$$

and so the quasi-linear diffusion equation conserves the density $n=\int \mathrm{d}vf_0$.

Conservation of momentum

Examination of momentum conservation requires taking the first moment of Eq. (14.36),

$$\frac{\partial}{\partial t}(nmu)=m\int \mathrm{d}v\,v\frac{\partial}{\partial v}\left(D_{QL}\frac{\partial f_0}{\partial v}\right)=-m\int \mathrm{d}v\,D_{QL}\frac{\partial f_0}{\partial v}. \tag{14.45}$$

Using Eq. (14.37) this becomes

$$\begin{aligned}\frac{\partial}{\partial t}(nmu)&=-m\int \mathrm{d}v\,\frac{2\mathrm{i}e^2}{\varepsilon_0 m^2}\int \mathrm{d}k\,\frac{\mathcal{E}(k,t)}{\omega-kv}\frac{\partial f_0}{\partial v}\\&=-2\mathrm{i}\omega_p^2\int \mathrm{d}k\mathcal{E}(k,t)\int \mathrm{d}v\frac{1}{\omega-kv}\frac{\partial \bar{f}_0}{\partial v}.\end{aligned} \tag{14.46}$$

However, the linear dispersion relation Eq. (14.42) implies

$$\omega_p^2\int \mathrm{d}v\frac{1}{\omega-kv}\frac{\partial \bar{f}_0}{\partial v}=-k \tag{14.47}$$

and so

$$\frac{\partial}{\partial t}(nmu)=2\mathrm{i}\int \mathrm{d}k\mathcal{E}(k,t)k=0, \tag{14.48}$$

which vanishes because the integrand is an odd function of k. Thus, the constraint provided by the linear dispersion relation shows that the quasi-linear velocity diffusion equation also conserves momentum.

Conservation of energy

Consideration of energy conservation starts out in a similar manner but leads to some interesting, non-trivial results. The mean particle kinetic energy is defined to be

$$W_P = \int \mathrm{d}v \frac{mv^2}{2} f_0. \tag{14.49}$$

The time evolution of W_P is obtained by taking the second moment of the quasi-linear diffusion equation, Eq. (14.36),

$$\begin{aligned}\frac{\partial W_P}{\partial t} &= \int \mathrm{d}v \frac{mv^2}{2} \frac{\partial}{\partial v} D_{QL} \frac{\partial f_0}{\partial v} \\ &= -\int \mathrm{d}v\, mv D_{QL} \frac{\partial f_0}{\partial v}.\end{aligned} \tag{14.50}$$

Using Eq. (14.37) gives

$$\begin{aligned}\frac{\partial W_P}{\partial t} &= -\int \mathrm{d}v\, mv \frac{2\mathrm{i}e^2}{\varepsilon_0 m^2} \int \mathrm{d}k\, \frac{\mathcal{E}(k,t)}{\omega - kv} \frac{\partial f_0}{\partial v} \\ &= 2\mathrm{i}\omega_p^2 \int \mathrm{d}k\, \frac{\mathcal{E}(k,t)}{k} \int \mathrm{d}v\, \frac{\omega - kv - \omega}{\omega - kv} \frac{\partial \bar{f}_0}{\partial v} \\ &= -\omega_p^2 \int \mathrm{d}k\, \frac{2\mathrm{i}\omega(k)\mathcal{E}(k,t)}{k} \int \mathrm{d}v\, \frac{1}{\omega - kv} \frac{\partial \bar{f}_0}{\partial v}.\end{aligned} \tag{14.51}$$

Invoking Eq. (14.43) to substitute for the velocity integral results in

$$\frac{\partial W_P}{\partial t} = \int \mathrm{d}k\, 2\mathrm{i}\left[\omega_r(k) + \mathrm{i}\omega_i(k)\right] \mathcal{E}(k,t). \tag{14.52}$$

Since $\omega_r(k)$ is an odd function of k and $\omega_i(k)$ is an even function of k, only the term involving ω_i survives the k integration and so

$$\frac{\partial W_P}{\partial t} + \int \mathrm{d}k\, 2\omega_i(k)\mathcal{E}(k,t) = 0. \tag{14.53}$$

Using Eq. (14.40) this becomes

$$\frac{\partial}{\partial t}\left[W_P + \int \mathrm{d}k\, \mathcal{E}(k,t)\right] = 0, \tag{14.54}$$

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which can now be integrated to give

$$W_P + \int \mathrm{d}k\, \mathcal{E}(k,t) = W_P + W_E = const.$$

showing that the sum of the particle and electric field energies is conserved. The particle energy and the electric field energy therefore need not be individually conserved – only the sum of these two types of energy is conserved. This result allows for energy exchange between the particles and the electric field.

14.2.3 Energy exchange with resonant particles

More detailed insight is obtained by considering the role of resonant particles, i.e., those particles having velocity $v \approx \omega/k$ as indicated by the shaded region in Fig. 14.1. This is done by using Eq. (14.38) to rewrite the top line of Eq. (14.51) as

$$\begin{aligned}\frac{\partial W_P}{\partial t} &= -\frac{2e^2}{\varepsilon_0 m}\int \mathrm{d}k\, \mathcal{E}(k,t)\int \mathrm{d}v\, v\frac{\mathrm{i}\,[\omega_r(k)-kv]+\omega_i(k)}{[\omega_r(k)-kv]^2+\omega_i^2(k)}\frac{\partial f_0}{\partial v} \\ &= -2\omega_p^2\int \mathrm{d}k\, \mathcal{E}(k,t)\int \mathrm{d}v\frac{v\omega_i(k)}{[\omega_r(k)-kv]^2+\omega_i^2(k)}\frac{\partial \bar{f}_0}{\partial v},\end{aligned} \tag{14.55}$$

where, because $\omega_r(k)-kv$ is an odd function of k, only the $\omega_i(k)$ numerator term survives the k integration. The velocity integral can be decomposed into a resonant portion, which is the velocity range where $\omega_r \simeq kv$, and the remaining or non-resonant portion. In the resonant portion, it is possible to approximate

$$\frac{\omega_i}{(\omega_r-kv)^2+\omega_i^2} \simeq \pi\delta(\omega_r-kv) = \frac{\pi}{k}\delta(v-\frac{\omega_r}{k}) \tag{14.56}$$

while the non-resonant portion can be written as a principle-part integral. Thus, Eq. (14.55) becomes

$$\frac{\partial W_P}{\partial t} = -\omega_p^2\int \mathrm{d}k\, 2\mathcal{E}(k,t)\left[P\int \mathrm{d}v\,\frac{v\omega_i(k)}{(\omega_r-kv)^2}\frac{\partial \bar{f}_0}{\partial v}+\pi\frac{\omega_r}{k^2}\left(\frac{\partial \bar{f}_0}{\partial v}\right)_{v=\omega_r/k}\right]. \tag{14.57}$$

A relationship between the principle part and resonant terms in this expression can be constructed by similarly decomposing Eq. (14.43) into a resonant portion and a non-resonant or principle part. The principle part is Taylor expanded as a function of $\omega_r + \mathrm{i}\omega_i$, where ω_i is assumed to be much smaller than ω_r. This procedure is essentially the scheme discussed in the development of Eq. (5.85), except that here the expansion of the principle part is written explicitly in order to emphasize certain details. Thus, the linear dispersion relation Eq. (14.43) can be expanded as

$$
\begin{aligned}
0 &= 1 - \frac{\omega_p^2}{k^2}\int \mathrm{d}v \frac{\partial \bar{f}_0/\partial v}{v - \dfrac{\omega_r}{k} - \dfrac{\mathrm{i}\omega_i}{k}} \\
&= 1 - \frac{\omega_p^2}{k^2}\left[P\int \mathrm{d}v \frac{\partial \bar{f}_0/\partial v}{v - \dfrac{\omega_r}{k} - \dfrac{\mathrm{i}\omega_i}{k}} + \mathrm{i}\pi\left(\frac{\partial \bar{f}_0}{\partial v}\right)_{v=\omega_r/k}\right] \\
&= 1 - \frac{\omega_p^2}{k^2}\left[P\int \mathrm{d}v \frac{\partial \bar{f}_0/\partial v}{v - \dfrac{\omega_r}{k}} + \frac{\mathrm{i}\omega_i}{k}\frac{\partial}{\partial(\frac{\omega}{k})}P\int \mathrm{d}v \frac{\partial \bar{f}_0/\partial v}{v - \dfrac{\omega}{k}} + \mathrm{i}\pi\left(\frac{\partial \bar{f}_0}{\partial v}\right)_{v=\omega_r/k}\right] \\
&= 1 - \frac{\omega_p^2}{k^2}\left[P\int \mathrm{d}v \frac{\partial \bar{f}_0/\partial v}{v - \dfrac{\omega_r}{k}} + \frac{\mathrm{i}\omega_i}{k}P\int \mathrm{d}v \frac{\partial \bar{f}_0/\partial v}{\left(v - \dfrac{\omega}{k}\right)^2} + \mathrm{i}\pi\left(\frac{\partial \bar{f}_0}{\partial v}\right)_{v=\omega_r/k}\right].
\end{aligned}
\tag{14.58}
$$

The imaginary part of the last line must vanish and so

$$
\frac{\omega_i}{k}P\int \mathrm{d}v \frac{\partial \bar{f}_0/\partial v}{(v - \omega_r/k)^2} + \pi\left(\frac{\partial \bar{f}_0}{\partial v}\right)_{v=\omega_r/k} = 0, \tag{14.59}
$$

which leads to the usual expression for Landau damping.

If it is assumed that $\omega_r/k \gg v_T$ then the principle-part integral in Eq. (14.57) can be approximated as

$$
P\int \mathrm{d}v \, \frac{v}{(\omega_r - kv)^2}\frac{\partial \bar{f}_0}{\partial v} \simeq \frac{1}{\omega_r^2}\int \mathrm{d}v \, v\frac{\partial \bar{f}_0}{\partial v} = -\frac{1}{\omega_r^2}\int \mathrm{d}v \bar{f}_0 = -\frac{1}{\omega_r^2} \tag{14.60}
$$

and the principle-part integral in Eq. (14.59) can similarly be approximated as

$$
\begin{aligned}
P\int \mathrm{d}v \frac{\partial \bar{f}_0/\partial v}{(v - \omega_r/k)^2} &= -\int \mathrm{d}v \bar{f}_0 \frac{\partial}{\partial v}\frac{1}{(v - \omega_r/k)^2} \\
&= 2\int \mathrm{d}v \frac{\bar{f}_0}{(v - \omega_r/k)^3} \\
&\simeq -2\frac{k^3}{\omega_r^3}.
\end{aligned}
\tag{14.61}
$$

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Using Eq. (14.60), Eq. (14.57) becomes

$$\frac{\partial W_P}{\partial t} = -\omega_p^2 \int \mathrm{d}k\, 2\mathcal{E}(k,t)\left[-\frac{\omega_i(k)}{\omega_r^2} + \frac{\omega_r}{k^2}\pi\left(\frac{\partial \bar{f}_0}{\partial v}\right)_{v=\omega_r/k}\right] \tag{14.62}$$

and using Eq. (14.61), Eq. (14.59) becomes

$$-2\omega_i\frac{k^2}{\omega_r^3} + \pi\left(\frac{\partial \bar{f}_0}{\partial v}\right)_{v=\omega_r/k} = 0. \tag{14.63}$$

Comparison of the above two expressions shows that the second term in the square brackets of Eq. (14.62) has twice the magnitude of the first term and is of the opposite sign. Since $\omega_r^2 \simeq \omega_p^2$, this means Eq. (14.62) is of the form

$$\frac{\partial W_P}{\partial t} = \frac{\partial}{\partial t}W_{P,non-resonant} + \frac{\partial}{\partial t}W_{P,resonant} = \int \mathrm{d}k\mathcal{E}(k,t)\left[2\omega_i - 4\omega_i\right], \tag{14.64}$$

where the $2\omega_i$ term prescribes the rate of change of the kinetic energy of the non-resonant particles and the $-4\omega_i$ term prescribes the rate of change of the kinetic energy of the resonant particles. On the other hand, Eq. (14.53) showed that the rate of change of the total particle kinetic energy was equal and opposite to the total rate of change of the electric field energy. These two statements can be reconciled by asserting that the wave energy consists of equal parts of non-resonant particle kinetic energy and electric field energy and that the resonant particles act as a source or sink for this wave energy. This energy budgeting is shown schematically as

$$\underbrace{\underbrace{\int \mathrm{d}k 2\omega_i(k)\mathcal{E}(k,t)}_{\substack{\text{kinetic energy}\\ \text{of non-resonant particles}}} + \underbrace{\int \mathrm{d}k 2\omega_i(k)\mathcal{E}(k,t)}_{\substack{\text{energy stored in}\\ \text{electric field}}}}_{\substack{\text{wave}\\ \text{energy}}} \Longleftrightarrow \underbrace{\int \mathrm{d}k 4\omega_i(k)\mathcal{E}(k,t)}_{\substack{\text{kinetic energy}\\ \text{of resonant particles}}}. \tag{14.65}$$

Behavior of the resonant particles

We define f_{res} as the velocity distribution of the resonant particles, i.e., the particles with velocities $v \simeq \omega/k$ for which $\mathcal{E}(k,t)$ is finite. Since the velocity range $v = \omega(k)/k$ of the resonant particles maps to the spectrum $\mathcal{E}(k,\, t)$, the upper and lower bounds of the resonant particle velocity range respectively map to the lower and upper bounds of the values of k for which $\mathcal{E}(k,\, t)$ is finite. For these

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particles, the delta function approximation, Eq. (14.56), can be used to evaluate the quasi-linear diffusion coefficient given by Eq. (14.39) and obtain

$$\begin{aligned} D_{QL,res}(v) &\simeq \frac{2\pi e^2}{\varepsilon_0 m^2}\int \mathrm{d}k\,\delta(\omega_r - kv)\mathcal{E}(k,t) \\ &= \frac{2\pi e^2}{\varepsilon_0 m^2}\int \mathrm{d}k\,\delta(\omega_r/v - k)\frac{\mathcal{E}(k,t)}{v} \\ &= \frac{2\pi e^2}{\varepsilon_0 m^2}\frac{\mathcal{E}(\omega_r/v,t)}{v}. \end{aligned} \tag{14.66}$$

Using this coefficient the quasi-linear velocity diffusion for the resonant particles is

$$\frac{\partial f_{0,res}}{\partial t} = \frac{2\pi e^2}{\varepsilon_0 m^2}\frac{\partial}{\partial v}\left(\frac{\mathcal{E}(\omega_r/v,\,t)}{v}\frac{\partial f_{0,res}}{\partial v}\right). \tag{14.67}$$

It is seen from Eq. (14.63), the generalized formula for Landau damping, that

$$\pi\left(\frac{\partial f_0}{\partial v}\right)_{v=\omega_r/k} = 2\omega_i\frac{k^2}{\omega_r^3}n_0 \tag{14.68}$$

and so Eq. (14.67) becomes

$$\begin{aligned} \frac{\partial f_{0,res}}{\partial t} &= \frac{2e^2}{\varepsilon_0 m^2}\frac{\partial}{\partial v}\left(\frac{\mathcal{E}(\omega_r/v,t)}{v}2\omega_i\frac{k^2}{\omega_r^3}n_0\right) \\ &\simeq \frac{2\omega_p}{m}\frac{\partial}{\partial v}\left(\frac{\mathcal{E}(\omega_r/v,t)}{v^3}2\omega_i\right) \\ &= \frac{2\omega_p}{m}\frac{\partial}{\partial t}\frac{\partial}{\partial v}\left(\frac{\mathcal{E}(\omega_r/v,t)}{v^3}\right), \end{aligned} \tag{14.69}$$

where $\omega_r \simeq \omega_p = \sqrt{n_0 e^2/\varepsilon_0 m}$ has been used and also, because the particles are resonant, $v \simeq \omega_r/k$. This can be integrated with respect to time to obtain

$$f_{0,res}(v,\,t) - f_{0,res}(v,\,0) = \frac{2\omega_p}{m}\frac{\partial}{\partial v}\left(\frac{\mathcal{E}(\omega_r/v,t) - \mathcal{E}(\omega_r/v,0)}{v^3}\right). \tag{14.70}$$

By definition $\mathcal{E}(\omega_r/v,t)$ vanishes for v lying outside the resonant particle velocity range $v_{res}^{\min} < v < v_{res}^{\max}$ because outside this range there is no wave energy with which the particles can resonate (see Fig. 14.1). Hence, integration of Eq. (14.70) over the velocity range $v_{res}^{\min} < v < v_{res}^{\max}$ of the resonant particles gives

$$\int_{v_{res}^{\min}}^{v_{res}^{\max}} \mathrm{d}v\, f_{0,res}(v,\,t) = \int_{v_{res}^{\min}}^{v_{res}^{\max}} \mathrm{d}v\, f_{0,res}(v,\,0), \tag{14.71}$$

which demonstrates that the number of resonant particles is conserved.

Equation (14.65) showed that the resonant particle energy is not conserved and can be exchanged with the wave energy. Thus, the zeroth moment of the resonant particles is conserved, but the second moment is not. Change of the resonant particle energy while conserving the total number of resonant particles is achieved by simultaneously adjusting the number of resonant particles that are slightly slower than ω/k and the number of resonant particles that are slightly faster. For example, if there is a decrease in the number of resonant particles having $v \simeq (\omega/k)_-$ there must be a corresponding increase in the number of resonant particles having $v \simeq (\omega/k)_+$. This process provides a net transfer of energy from the wave to the resonant particles, involves wave Landau damping, and requires having $\partial f_{0,res}/\partial v < 0$. The result, increasing the number of resonant particles having $v \simeq (\omega/k)_+$ while decreasing the number having $v \simeq (\omega/k)_-$, flattens $f_{0,res}(v)$ and so makes it plateau-like as indicated in Fig. 14.1(b).

Behavior of the non-resonant particles

The quasi-linear diffusion coefficient for the non-resonant particles comes from the principle part of the integral in Eq. (14.39), i.e.,

$$D_{QL,non\text{-}res} = \frac{e^2}{\varepsilon_0 m^2} P \int \mathrm{d}k \, \frac{2\omega_i(k)\mathcal{E}(k,t)}{[\omega_r(k) - kv]^2 + \omega_i^2(k)}. \tag{14.72}$$

The vast majority of the non-resonant particles have velocities much slower than the wave, so for the non-resonant particles it can be assumed $v \ll \omega_r/k$, in which case

$$\begin{aligned} D_{QL,non\text{-}res} &\simeq \frac{e^2}{\varepsilon_0 m^2} \int \mathrm{d}k \, \frac{2\omega_i(k)\mathcal{E}(k,t)}{\omega_r^2(k)} \\ &\simeq \frac{1}{mn_0} \int \mathrm{d}k \, 2\omega_i(k)\mathcal{E}(k,t). \end{aligned} \tag{14.73}$$

Since this non-resonant particle velocity-space diffusion coefficient is velocity-independent, it can be factored from velocity integrals or derivatives. Equation (14.12) showed that the change in f_0 is order ϵ^2, where $\epsilon \ll 1$. This means that changes in f_0 are small compared to f_0. On the other hand, there is no zero-order wave energy since the wave energy scales as E_1^2 and so is entirely constituted of terms that are order ϵ^2. Thus, the wave-energy spectrum can change substantially (for example, disappear altogether) whereas there is a only slight corresponding change to f_0. Thus, Eq. (14.36), becomes

$$\begin{aligned} \frac{\partial f_{0,non\text{-}res}}{\partial t} &= \frac{1}{mn_0} \frac{\partial}{\partial v} \int \mathrm{d}k \, 2\omega_i(k)\mathcal{E}(k,t) \frac{\partial f_{0,non-res}}{\partial v} \\ &\simeq \frac{1}{mn_0} \left(\frac{\mathrm{d}}{\mathrm{d}t} \int \mathrm{d}k \, \mathcal{E}(k,t) \right) \frac{\partial^2 f_{0,non-res}}{\partial v^2}, \end{aligned} \tag{14.74}$$

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which can be integrated to give

$$f_{0,non\text{-}res}(v,t) - f_{0,non\text{-}res}(v,0) = \frac{1}{mn_0}\left(\int \mathrm{d}k\ [\mathcal{E}(k,t) - \mathcal{E}(k,0)]\right)\frac{\partial^2 f_{0,non\text{-}res}}{\partial v^2}. \tag{14.75}$$

Since the number of resonant particles is conserved, the number of non-resonant particles must also be conserved.

If an initial wave spectrum becomes damped at $t = \infty$ then it is possible to write

$$f_{0,non\text{-}res}(v,\infty) = f_{0,non\text{-}res}(v,0) - \frac{1}{mn_0}\left[\int \mathrm{d}k\, \mathcal{E}(k,0)\right]\frac{\partial^2 f_{0,non\text{-}res}(v,0)}{\partial v^2}. \tag{14.76}$$

In the range of velocities where $\partial^2 f_{0,non\text{-}res}/\partial v^2 > 0$ there will be a decrease in the number of non-resonant particles and vice versa in the range where $\partial^2 f_{0,non\text{-}res}/\partial v^2 < 0$. An initially Maxwellian distribution (which has $\partial^2 f_{0,non\text{-}res}/\partial v^2 < 0$ for very small velocities and vice versa for very large velocities) will develop a double-plateau shape (high plateau at low velocities and low plateau at high velocities) with a sharp gradient between the two plateaus. The lower plateau will merge smoothly with the plateau of the resonant particles. The non-resonant portion of the velocity distribution will appear to become colder as the waves damp as indicated in Fig. 14.1(b).

This apparent cooling can be quantified by introducing an effective temperature, T_{eff}, which is defined to have the time derivative

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\kappa T_{eff}\right) = \frac{2}{n_0}\frac{\mathrm{d}}{\mathrm{d}t}\int \mathrm{d}k\, \mathcal{E}(k,t). \tag{14.77}$$

This expression can be integrated to obtain

$$\kappa T_{eff}(t) = \kappa T_0 + \frac{2}{n_0}\int \mathrm{d}k\mathcal{E}(k,t), \tag{14.78}$$

where T_0 is the temperature for the situation where there are no waves.

Using Eq. (14.77), Eq. (14.74) can be written as

$$\frac{\partial}{\partial t} f_{0,non\text{-}res} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\kappa T_{eff}}{2m}\right)\frac{\partial^2 f_{0,non\text{-}res}}{\partial v^2}. \tag{14.79}$$

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If $f_{0,non\text{-}res}$ is considered a function of κT_{eff} instead of t, Eq. (14.79) can be written as

$$\frac{\partial}{\partial(\kappa T_{eff})} f_{0,non\text{-}res} = \frac{1}{2m} \frac{\partial^2 f_{0,non\text{-}res}}{\partial v^2}, \tag{14.80}$$

which has the appropriately normalized solution

$$\begin{aligned} f_{0,non\text{-}res} &= n_0 \sqrt{\frac{m}{2\pi\kappa T_{eff}(t)}} \exp\left(-\frac{mv^2}{2\kappa T_{eff}(t)}\right) \\ &= n_0 \sqrt{\frac{m/2\pi}{\kappa T_0 + \frac{2}{n_0}\int \mathrm{d}k\, \mathcal{E}(k,t)}} \exp\left(-\frac{mv^2/2}{\kappa T_0 + \frac{2}{n_0}\int \mathrm{d}k\, \mathcal{E}(k,t)}\right). \end{aligned} \tag{14.81}$$

Thus, wave damping (i.e., the reduction of $\mathcal{E}(k,t)$) corresponds to an effective cooling of the non-resonant particles. As shown in Eq. (14.65) this kinetic energy reduction is accompanied by an equal reduction in the electric field energy and all this energy is transferred to the resonant particles.

14.3 Echoes

Plasma wave echoes (Gould, O'Neil, and Malmberg 1967, Malmberg *et al.* 1968) are a nonlinear effect whereby (i) a first wave is excited and then dies out from Landau damping, (ii) a second wave is excited and similarly dies out, and then finally (iii) at a much later time (or much more distant location), long after the first two waves have disappeared, a ghost-like third wave appears, seemingly from nowhere but with properties depending on the first two waves. Echoes provide useful insights into the Landau damping mechanism and also raise interesting questions about how Landau damping relates to entropy. These questions arise from consideration of the following apparently conflicting observations:

1. Wave damping should destroy the information content of the wave by converting ordered motion into heat. Thus, wave damping should increase the entropy of the system.
2. The collisionless Vlasov equation conserves entropy. This is because collisions are the agent that increases randomness and hence entropy. A collisionless system is in principle completely deterministic, so the future of the system can be predicted with complete precision simply by integrating the system of equations forward in time. The

entropy-conserving property of the collisionless Vlasov equation can be verified by direct calculation of the rate of change of the entropy,

$$\begin{aligned}
\frac{\mathrm{d}S}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\int \mathrm{d}x \int \mathrm{d}v\, f(x,v,t)\ln f(x,v,t) \\
&= \int \mathrm{d}x \int \mathrm{d}v\, (\ln f + 1)\,\frac{\partial f}{\partial t} \\
&= -\int \mathrm{d}x \int \mathrm{d}v\, (\ln f + 1)\left(v\frac{\partial f}{\partial x} + \frac{q}{m}E\frac{\partial f}{\partial v}\right) \\
&= -\int \mathrm{d}x \int \mathrm{d}v\, \ln f\left(v\frac{\partial f}{\partial x} + \frac{q}{m}E\frac{\partial f}{\partial v}\right) \\
&= -\int \mathrm{d}x \int \mathrm{d}v\, \left(v\frac{\partial}{\partial x} + \frac{q}{m}E\frac{\partial}{\partial v}\right)(f\ln f - f) \\
&= 0
\end{aligned} \tag{14.82}$$

since

$$\begin{gathered}
\int \mathrm{d}v\frac{\partial f}{\partial v} = 0, \quad \int \mathrm{d}x\frac{\partial f}{\partial x} = 0, \\
\int \mathrm{d}v\frac{\partial}{\partial v}(f\ln f - f) = 0, \quad \int \mathrm{d}x\frac{\partial}{\partial x}(f\ln f - f) = 0.
\end{gathered} \tag{14.83}$$

So, what really happens – does Landau damping increase entropy or not? The answer goes right to the heart of what is meant by entropy. In particular, it should be recalled that entropy is defined as the natural logarithm of the number of microscopic states corresponding to a given macroscopic state (see Section. 2.5.1). The concept "macroscopic state" implies the existence of a statistically large number of microscopic states that are indistinguishable from each other for all intents and purposes using all available means of observation. Collisions would cause the system to continuously evolve through all the various microscopic states and an observer of the system would not be able to distinguish one microscopic state from another, even if equipped with the best available measuring equipment. The system could then evolve through the various microscopic states that map to a specific macroscopic state and the observer would not notice.

The paradox of whether Landau damping increases entropy or not is resolved because the concept of many microscopic states mapping to a single macroscopic state is invalid for Landau damping. In actual fact, the physical system for the Landau damping problem is in just one well-defined, calculable state. The macroscopic state therefore maps to just one microscopic state and so the system does not continuously and randomly evolve through a sequence of microscopic states.

Landau damping does not involve turning ordered information into heat. Instead, macroscopically ordered information is turned into microscopically ordered information. The information still exists, but is encoded in a macroscopically invisible form. A good analogy is the process of making a holographic image of an object. The hologram looks blank unless illuminated in a special way that provides the proper decoding. If an observer does not know how to do the decoding, the information is lost to the observer, and the entropy content of the hologram appears high compared to a regular photograph. However, if the observer knows how to decode the microscopically stored information, then no information is lost.

The Landau damping process scrambles the phase of the macroscopic information and encodes this as microscopic information, which is normally irretrievable. If one does not know the appropriate trick for unscrambling the microscopically encoded information, it would be tempting to state that entropy has increased. However, the appropriate trick exists and has been demonstrated in laboratory experiments. Information that appeared to have been lost is retrieved and so entropy is not increased.

To get an idea for the issues involved, we first consider the simple problem of a one-dimensional beam of electrons having initial velocity v_0. The beam is transiently accelerated by an externally generated, spatially periodic electric field pulse $E = \bar{E}\cos(kx)\delta(t)$, where $\bar{E}$ is considered to be infinitesimal. The equation of motion for the electron beam is

$$m\left(\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x}\right) = -e\bar{E}\cos(kx)\delta(t) \tag{14.84}$$

and linearization of this equation gives

$$m\left(\frac{\partial v_1}{\partial t} + v_0\frac{\partial v_1}{\partial x}\right) = -e\bar{E}\cos(kx)\delta(t). \tag{14.85}$$

It is convenient to use complex notation so that $\cos kx \to \mathrm{e}^{\mathrm{i}kx}$ and it is understood that, eventually, the real part of a complex solution will be used to give the physical solution. It is therefore assumed that $v_1 \sim \mathrm{e}^{\mathrm{i}kx}$ and so the linearized equation becomes

$$m\left(\frac{\partial v_1}{\partial t} + \mathrm{i}kv_0v_1\right) = -e\bar{E}\delta(t). \tag{14.86}$$

Next it is argued that since the delta function can be considered as the superposition of an infinite spectrum of harmonic oscillations, i.e.,

$$\delta(t) = \frac{1}{2\pi}\int \mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}\omega, \tag{14.87}$$

the response of the beam to each frequency component can be considered. Thus, we consider the equation

$$\frac{\partial \tilde{v}_1}{\partial t} + \mathrm{i}kv_0\tilde{v}_1 = -\frac{e\bar{E}}{m}\mathrm{e}^{-\mathrm{i}\omega t}, \tag{14.88}$$

so the net velocity is

$$v_1(t) = \frac{1}{2\pi}\int \tilde{v}_1(\omega, t)\mathrm{d}\omega. \tag{14.89}$$

Note that $\tilde{v}_1(\omega, t)$ is *not* a Fourier transform because it contains the explicit time dependence.

Since v_0 is the initial beam velocity, $v_1(t)$ must vanish at $t = 0$ providing a boundary condition for Eq. (14.88) at $t = 0$. One way to solve this equation is first to assume that $\tilde{v}_1(\omega, t)$ consists of a particular solution satisfying the inhomogeneous part of the equation (i.e., balances the driving term on the right-hand side) and a homogeneous solution (i.e., a solution of the homogeneous equation [left-hand side of Eq. (14.88)]). The coefficient of the homogeneous solution is chosen to satisfy the boundary condition at $t = 0$. The particular solution is assumed to vary as $\mathrm{e}^{\mathrm{i}kx-\mathrm{i}\omega t}$ and so is the solution of the equation

$$(-\mathrm{i}\omega + \mathrm{i}kv_0)\,\tilde{v}_1 = -\frac{e\bar{E}}{m}\mathrm{e}^{-\mathrm{i}\omega t}. \tag{14.90}$$

The homogenous solution is the solution of

$$\frac{\partial \tilde{v}_1}{\partial t} + \mathrm{i}kv_0\tilde{v}_1 = 0 \tag{14.91}$$

and has the form $\tilde{v}_1^h = \lambda \exp(-\mathrm{i}kv_0 t)$, where λ is a constant to be determined. Adding the particular and homogeneous solutions together gives the general solution

$$\tilde{v}_1 = -\frac{e\bar{E}}{m}\frac{\mathrm{i}e^{-\mathrm{i}\omega t}}{(\omega - kv_0)} + \lambda \mathrm{e}^{-\mathrm{i}kv_0 t}, \tag{14.92}$$

where λ is chosen to satisfy the initial condition. The initial condition $v_1 = 0$ at $t = 0$ determines λ and gives

$$\tilde{v}_1(\omega, t) = -\frac{\mathrm{i}e\bar{E}}{m}\frac{\left(\mathrm{e}^{-\mathrm{i}\omega t} - \mathrm{e}^{-\mathrm{i}kv_0 t}\right)}{(\omega - kv_0)} \tag{14.93}$$

as the solution that satisfies both Eq. (14.88) and the initial condition. The term involving $\mathrm{e}^{-\mathrm{i}kv_0 t}$ is called the ballistic term. This term contains information about the initial conditions, is a solution of the homogeneous equation, is missed by Fourier treatments, is incorporated by Laplace transform treatments, and keeps v_1 from diverging when $\omega - kv_0 \to 0$.

If we wished to revert to the time domain, then the contributions of all the harmonics would have to be summed, giving

$$v_1(t) = -\frac{\mathrm{i}e\bar{E}}{2\pi m}\int \mathrm{d}\omega \frac{\left(\mathrm{e}^{-\mathrm{i}\omega t} - \mathrm{e}^{-\mathrm{i}kv_0 t}\right)}{(\omega - kv_0)}. \tag{14.94}$$

14.3.1 Ballistic terms and Laplace transforms

The discussion above used an approach related to Fourier transforms, but added additional structure to account for the initial condition $v_1 = 0$ at $t = 0$. This suggests Laplace transforms ought to be used, since Laplace transforms automatically take into account initial conditions. Let us therefore Laplace transform Eq. (14.88) to see if indeed the particular and ballistic terms are appropriately characterized. The Laplace transform of Eq. (14.88) gives

$$\begin{aligned} p\tilde{v}_1 + \mathrm{i}kv_0\tilde{v}_1 &= -\frac{e\bar{E}}{m}\int_0^\infty \mathrm{d}t\,\mathrm{e}^{-\mathrm{i}\omega t - pt} \\ &= -\frac{e\bar{E}}{m}\frac{1}{\mathrm{i}\omega + p} \end{aligned} \tag{14.95}$$

so

$$\tilde{v}_1 = -\frac{e\bar{E}/m}{(p+\mathrm{i}\omega)(p+\mathrm{i}kv_0)}. \tag{14.96}$$

The inverse Laplace transform gives

$$\tilde{v}_1 = -\frac{1}{2\pi\mathrm{i}}\frac{e\bar{E}}{m}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty} \frac{\mathrm{e}^{pt}}{(p+\mathrm{i}\omega)(p+\mathrm{i}kv_0)}\mathrm{d}p. \tag{14.97}$$

Analytic continuation allows the contour to be completed on the left-hand side and it is seen that there are two poles, one at $p = -\mathrm{i}\omega$ and the other at $p = -\mathrm{i}kv$. Evaluation of the residues for the two poles gives

$$\begin{aligned} \tilde{v}_1 &= -\frac{1}{2\pi\mathrm{i}}\frac{e\bar{E}}{m}\left\{\begin{array}{l} 2\pi\mathrm{i}\lim\limits_{p\to-\mathrm{i}\omega}(p+\mathrm{i}\omega)\left[\dfrac{\mathrm{e}^{pt}}{(p+\mathrm{i}\omega)(p+\mathrm{i}kv_0)}\right] + \\ 2\pi\mathrm{i}\lim\limits_{p\to-\mathrm{i}kv}(p+\mathrm{i}kv)\left[\dfrac{\mathrm{e}^{pt}}{(p+\mathrm{i}\omega)(p+\mathrm{i}kv_0)}\right] \end{array}\right\} \\ &= -\frac{e\bar{E}}{m}\left\{\frac{\mathrm{e}^{-\mathrm{i}\omega t}}{(-\mathrm{i}\omega + \mathrm{i}kv_0)} + \frac{\mathrm{e}^{-\mathrm{i}kv_0 t}}{(-\mathrm{i}kv_0 + \mathrm{i}\omega)}\right\}, \end{aligned} \tag{14.98}$$

which is the same as Eq. (14.93). The ballistic term $\sim \mathrm{e}^{-\mathrm{i}kv_0 t}$ thus results from a pole originating from the Laplace transform of the source term, whereas the homogeneous term $\sim \mathrm{e}^{-\mathrm{i}\omega t}$ results from the pole originating from the factor $(p + \mathrm{i}kv_0)$ appearing on the left-hand side of Eq. (14.95).

14.3.2 Phase mixing of the ballistic term for multiple beams

Now suppose that instead of just one electron beam, there are multiple beams where the density of each beam is proportional to $\exp\left(-v_0^2/v_T^2\right)$; this would be one way of characterizing a Maxwellian velocity distribution. Superposition of the ballistic terms from all these beams leads to a vanishing sum, because the ballistic terms each have a velocity-dependent phase and so the superposition would give destructive interference due to phase mixing. In particular, the superposition would involve integrals of the form

$$\int \mathrm{d}v_0 \mathrm{e}^{-v_0^2/v_T^2+\mathrm{i}kv_0t} = \mathrm{e}^{-k^2v_T^2t^2/4} \int \mathrm{d}v_0 \mathrm{e}^{-(v_0/v_T+\mathrm{i}kv_Tt/2)^2} = \sqrt{\pi} v_T \mathrm{e}^{-k^2v_T^2t^2/4}, \tag{14.99}$$

which would quickly become extremely small. This suggests the ballistic term has no enduring macroscopic physical importance and, in fact, this is true for linear problems where the ballistic term is typically ignored. However, the ballistic term can assume importance when nonlinearities are considered.

14.3.3 Beam echoes

The linearized continuity equation corresponding to Eq. (14.85) is

$$\frac{\partial n_1}{\partial t} + v_0 \frac{\partial n_1}{\partial x} + n_0 \frac{\partial v_1}{\partial x} = 0 \tag{14.100}$$

and so, invoking the assumed $\mathrm{e}^{\mathrm{i}kx}$ dependence, this becomes

$$\frac{\partial n_1}{\partial t} + \mathrm{i}kv_0 n_1 + \mathrm{i}kn_0 v_1 = 0. \tag{14.101}$$

In analogy to Eq. (14.89) the linearized density can be expressed as

$$n_1(t) = \frac{1}{2\pi} \int \tilde{n}_1(\omega, t)\mathrm{d}\omega, \tag{14.102}$$

where again $\tilde{n}_1(\omega, t)$ is not a Fourier transform because $\tilde{n}_1(\omega, t)$ contains a ballistic term incorporating information about initial conditions. The equation for each frequency component thus is

$$\frac{\partial \tilde{n}_1}{\partial t} + \mathrm{i}kv_0 \tilde{n}_1 + \mathrm{i}kn_0 \tilde{v}_1 = 0, \tag{14.103}$$

which may be Laplace transformed to give

$$(p + \mathrm{i}kv_0)\, \tilde{n}_1 + \mathrm{i}kn_0 \tilde{v}_1 = 0 \tag{14.104}$$

or, using Eq. (14.96),

$$\tilde{n}_1(p, \omega) = \frac{\mathrm{i}kn_0 e\bar{E}/m}{(p + \mathrm{i}\omega)\,(p + \mathrm{i}kv_0)^2}. \tag{14.105}$$

There is now a second-order pole at $p = \mathrm{i}kv_0$ and so there will be a ballistic term $\sim \exp(\mathrm{i}kv_0 t)$ associated with the density perturbation. This ballistic term will also phase mix away if there is a Gaussian velocity distribution of beams.

In order to consider nonlinear consequences, we consider the second-order continuity equation

$$\frac{\partial n_2}{\partial t} + v_0 \frac{\partial n_2}{\partial t} + v_1 \frac{\partial n_1}{\partial x} + \frac{\partial}{\partial x}(n_1 v_1) = 0, \tag{14.106}$$

which has inhomogeneous (forcing) terms such as $n_1 v_1$ involving products of linear quantities. Suppose that in addition to the original pulse with spatial wavenumber k at time $t = 0$ an additional pulse is also imposed with wavenumber k' at time $t = \tau$. This additional pulse would introduce ballistic terms having a time dependence $\sim \exp(\mathrm{i}k' v_0 (t - \tau))$. Thus, the nonlinear product $n_1 v_1$ includes a dependence

$$n_1 v_1 \sim \mathrm{Re}\left[\exp(\mathrm{i}kv_0 t)\right] \times \mathrm{Re}\left[\exp(\mathrm{i}k' v_0 (t - \tau))\right], \tag{14.107}$$

which contains terms of the form $\exp(\mathrm{i}kv_0 t - k' v_0 (t - \tau))$. In general, this product of ballistic terms would phase mix away if there were a Gaussian distribution of beams, just as for the linear ballistic term. However, at the special time given by

$$kt - k'(t - \tau) = 0, \tag{14.108}$$

the phase of the nonlinear ballistic term vanishes for *all* velocities, and so no phase mixing occurs when the velocity contributions are summed. Thus, at the special time

$$t = \frac{k' \tau}{k' - k}, \tag{14.109}$$

the nonlinear product is immune to phase mixing and the superposition of the nonlinear ballistic terms of a Gaussian distribution of beams gives a macroscopic signal. The time at which the phase of the nonlinear ballistic term becomes stationary can greatly exceed τ and so a macroscopic nonlinear signal appears long after both initial pulses have gone. This ghost-like nonlinear signal is called the fluid echo.

14.3.4 The self-consistent nonlinear Vlasov–Poisson problem

These ideas carry over into the Vlasov–Poisson analysis of plasma waves, but several additional issues occur that complicate and obscure matters. First, the problem takes place in phase-space so, instead of having a pair of coupled equations for velocity and density, there is only the Vlasov equation. The Vlasov equation not only contains a convective term analogous to the fluid convective terms but also incorporates an acceleration term, which describes how the velocity distribution function is modified when particles undergo acceleration and change their velocity. As in the fluid equations, there is a coupling with Poisson's equation. This coupling not only provides the self-consistent interaction giving plasma waves, but also provides Landau damping. Landau damping is essentially a phase mixing of the Fourier-like driven terms, each of which scales as $\int \mathrm{d}v(\omega - kv)^{-1} \exp(-\mathrm{i}\omega t)\partial f_0/\partial v$. However, linear ballistic terms must also be excited in order to satisfy initial conditions. These ballistic terms scale as $\int \mathrm{d}v(\omega - kv)^{-1} \exp(-\mathrm{i}kvt)\partial f_0/\partial v$ and also phase mix away because of the $\exp(-\mathrm{i}kvt)$ factor.

If there are two successive pulses, then the nonlinear ballistic term will again have a stationary phase (i.e., phase independent of velocity) at the special time given by Eq. (14.109). This time could be arranged to be long after the linear plasma responses to the two pulses have Landau-damped away so the echo would seem to appear from nowhere. Thus, the Vlasov–Poisson analysis contains essentially similar echo physics, but in addition has a self-consistent treatment of the plasma waves and their associated Landau damping.

To proceed with the Vlasov–Poisson analysis, we begin by considering Eq. (14.8) again, which is rewritten below for convenience

$$\begin{aligned}\frac{\partial}{\partial t}(f_0+f_1+f_2+\ldots)+v\frac{\partial}{\partial x}(f_0+f_1+f_2+\ldots)\\ -\frac{e}{m}(E_0+E_1+E_2+\ldots)\frac{\partial}{\partial v}(f_0+f_1+f_2+\ldots)=0.\end{aligned} \tag{14.110}$$

Equilibrium is defined as being the solution obtained by balancing all the zeroth order terms, i.e.,

$$\frac{\partial f_0}{\partial t}+v\frac{\partial f_0}{\partial x}-\frac{e}{m}E_0\frac{\partial f_0}{\partial v}=0. \tag{14.111}$$

This is trivially satisfied by having $E_0 = 0, \partial f_0/\partial t = 0, \partial f_0/\partial x = 0$, and $f_0(v)$ arbitrary; we will make these assumptions here. This equilibrium solution is then subtracted from Eq. (14.110) leaving

$$\begin{aligned}\frac{\partial}{\partial t}(f_1+f_2+\ldots)+v\frac{\partial}{\partial x}(f_1+f_2+\ldots)-\frac{e}{m}E_1\frac{\partial}{\partial v}(f_0+f_1+f_2+\ldots)\\ -\frac{e}{m}(E_2+\ldots)\frac{\partial}{\partial v}(f_0+f_1+\ldots)=0.\end{aligned} \tag{14.112}$$

The first-order solution is defined as the solution to

$$\frac{\partial}{\partial t}f_1+v\frac{\partial f_1}{\partial x}-\frac{e}{m}E_1\frac{\partial f_0}{\partial v}=0, \tag{14.113}$$

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the equation obtained by retaining all the first-order terms. The first-order solution is then subtracted from Eq. (14.112) and what remains are second and higher order terms. Dropping terms higher than second-order gives the second-order equation

$$\frac{\partial f_2}{\partial t}+v\frac{\partial f_2}{\partial x}-\frac{eE_2}{m}\frac{\partial f_0}{\partial v}-\frac{e}{m}E_1\frac{\partial f_1}{\partial v}=0. \tag{14.114}$$

The special case where $E_1 f_1$ has a zero frequency component was discussed in the previous section on quasi-linear diffusion and so here only the situation where $E_1 f_1$ has a finite frequency will be considered. Since the first-order system has been solved, the last term in Eq. (14.114) can be considered as a source term and so we rearrange the equation to be

$$\frac{\partial f_2}{\partial t}+v\frac{\partial f_2}{\partial x}-\frac{eE_2}{m}\frac{\partial f_0}{\partial v}=\frac{e}{m}\frac{\partial}{\partial v}\left(E_1 f_1\right), \tag{14.115}$$

where E_1 and f_1 are known and f_2 is to be determined. The specification that $f=f_0+f_1$ at $t=0$ provides the boundary condition $f_2=0$ at time $t=0$. Because Eq. (14.115) has both a self-consistent electric field contribution (E_2 term on the left-hand side) and a prescribed electric field term (E_1 term on the right-hand side), it has aspects of both the self-consistent problem and of the problem where the electric field is prescribed.

The linear problem

Suppose periodic grids are inserted into a plasma and the grids are transiently pulsed thereby creating the potential $\phi_{ext}(x,t)$. The charged particles will move in response to this applied potential and the resulting displacement produces a perturbation of the plasma charge density. Poisson's equation must therefore take into account both the charge density on the grid and the resulting plasma charge density and so has the form

$$\nabla^2\phi_1=-\frac{1}{\varepsilon_0}\left(\text{grid charge density } + \text{ plasma charge density}\right). \tag{14.116}$$

Since the grid charge density is related to the grid potential by

$$\nabla^2\phi_{ext}=-\frac{1}{\varepsilon_0}\left(\text{grid charge density}\right) \tag{14.117}$$

Poisson's equation can be recast as

$$\nabla^2\phi_1=\nabla^2\phi_{ext}+\frac{e}{\varepsilon_0}\int f_1 \mathrm{d}v. \tag{14.118}$$

A Fourier–Laplace transform operation is then applied to Eq. (14.118) to obtain

$$-k^2\tilde{\phi}_1=-k^2\tilde{\phi}_{ext}+\frac{e}{\varepsilon_0}\int \tilde{f}_1 \mathrm{d}v. \tag{14.119}$$

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However, using $E_1 = -\nabla\phi_1$, the Fourier–Laplace transform of Eq. (14.113) gives

$$\tilde{f}_1 = -\mathrm{i}\frac{e}{m}\frac{k\tilde{\phi}_1}{(p+\mathrm{i}kv)}\frac{\partial f_0}{\partial v} \tag{14.120}$$

so Eq. (14.119) becomes

$$\tilde{\phi}_1(p,k) = \frac{\tilde{\phi}_{ext}(p,k)}{\mathcal{D}(p,k)}, \tag{14.121}$$

where

$$\mathcal{D}(p,k) = 1 - \mathrm{i}\frac{e^2}{k^2 m\varepsilon_0}\int\frac{k}{p+\mathrm{i}kv}\frac{\partial f_0}{\partial v}\mathrm{d}v \tag{14.122}$$

is the usual self-consistent linear dielectric response function. Thus, the driven first-order velocity distribution function can be written as

$$\tilde{f}_1 = -\mathrm{i}\frac{e}{m}\frac{k}{(p+\mathrm{i}kv)}\frac{\tilde{\phi}_{ext}(p,k)}{\mathcal{D}(p,k)}\frac{\partial f_0}{\partial v}. \tag{14.123}$$

If an inverse transform were to be performed on $\tilde{f}_1$, then three types of contribution to the final result need to be taken into account. These contributions are (i) a ballistic term $\sim \exp(\mathrm{i}kvt)$ associated with the pole due to $p+\mathrm{i}kv$ in the denominator, (ii) a Landau-damped plasma oscillation due to the pole resulting from the root of $\mathcal{D}(p,k)$ in the denominator, and (iii) the direct inverse transform of the numerator $\tilde{\phi}_{ext}(p,k)$ modified by the presence of the other factors.

Fourier–Laplace transform of the non-linear equation

As discussed in Eq. (14.1), nonlinear quantities provide sum and difference frequencies. For example, an oscillation at frequency ω would result from the nonlinear product of a term oscillating at $\omega-\omega'$ and a term oscillating at ω' since $(\omega-\omega')+\omega' = \omega$. This can be written in a more formal and more general way using convolution integrals for both Fourier and Laplace transforms.

Since Laplace transforms will be used for the temporal dependence and since the right-hand side of Eq. (14.115) involves a product term, it is necessary to consider the Laplace transform of a product,

$$\begin{aligned}
\mathcal{L}\left(g(t)h(t)\right) &= \int_0^\infty g(t)h(t)\mathrm{e}^{-pt}\mathrm{d}t \\
&= \int_0^\infty \left[\frac{1}{2\pi\mathrm{i}}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\tilde{g}(p')\mathrm{e}^{p't}\mathrm{d}p'\right]h(t)\mathrm{e}^{-pt}\mathrm{d}t \\
&= \frac{1}{2\pi\mathrm{i}}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\tilde{g}(p')\left(\int_0^\infty h(t)\mathrm{e}^{(p'-p)t}\mathrm{d}t\right)\mathrm{d}p' \\
&= \frac{1}{2\pi\mathrm{i}}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\tilde{g}(p')\tilde{h}(p-p')\mathrm{d}p'.
\end{aligned} \tag{14.124}$$

This means that the Laplace transform with argument p results from all possible products of $\tilde{g}(p')$ with $\tilde{h}(p-p')$. If $\exp(\alpha_1 t)$ is the fastest-growing term in $g(t)$ and $\exp(\alpha_2 t)$ is the fastest growing term in $h(t)$, then $b > \alpha_1$ is required in order for the Laplace transform of $g(t)$ to be defined. Furthermore, in order for the Laplace transform of $h(t)$ to be defined, it is necessary to have $\mathrm{Re}[p] - \mathrm{Re}[p'] > \alpha_2$ or $\mathrm{Re}[p] > \alpha_2 + \mathrm{Re}[p']$, which implies $\mathrm{Re}[p] > b + \alpha_2$. These requirements can be summarized as $\mathrm{Re}[p] - \alpha_2 > b > \alpha_1$.

Using

$$\tilde{g}(k) = \int_{-\infty}^{\infty} g(x)\mathrm{e}^{-\mathrm{i}kx}\mathrm{d}x \tag{14.125a}$$

$$g(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{g}(k)\mathrm{e}^{+\mathrm{i}kx}\mathrm{d}k, \tag{14.125b}$$

a similar procedure for Fourier transforms gives

$$\begin{aligned}
\mathcal{F}(g(x)h(x)) &= \int_{-\infty}^{\infty} g(x)h(x)\mathrm{e}^{\mathrm{i}kx}\mathrm{d}x \\
&= \int_{-\infty}^{\infty}\left[\frac{1}{2\pi}\int_{-\infty}^{\infty}\tilde{g}(k')\mathrm{e}^{-\mathrm{i}k'x}\,\mathrm{d}k'\right]h(x)\mathrm{e}^{\mathrm{i}kx}\mathrm{d}x \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{d}k'\tilde{g}(k')\int_{-\infty}^{\infty}h(x)\mathrm{e}^{\mathrm{i}(k-k')x}\mathrm{d}x \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{d}k'\tilde{g}(k')\tilde{h}(k-k').
\end{aligned} \tag{14.126}$$

Thus, the Fourier–Laplace transform of Eq. (14.115) gives

$$\begin{aligned}
&(p+\mathrm{i}kv)\tilde{f}_2(p,k) + \mathrm{i}k\frac{e}{m}\frac{\partial f_0}{\partial v}\tilde{\phi}_2(p,k) \\
&= -\frac{e}{m}\frac{\partial}{\partial v}\left[\int_{-\infty}^{\infty}\frac{\mathrm{d}k'}{2\pi}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}k'\tilde{\phi}_1(p',k')\tilde{f}_1(p-p',k-k')\right].
\end{aligned} \tag{14.127}$$

The convolution integrals are notationally unwieldy and so to simplify the notation we define the new dummy variables

$$\begin{aligned}
\bar{k}' &= k - k' \\
\bar{p}' &= p - p',
\end{aligned} \tag{14.128}$$

in which case Eq. (14.127) becomes

$$\begin{aligned}
&(p+\mathrm{i}kv)\tilde{f}_2(p,k) + \mathrm{i}k\frac{e}{m}\frac{\partial f_0}{\partial v}\tilde{\phi}_2(p,k) \\
&= -\frac{e}{m}\frac{\partial}{\partial v}\left[\int_{-\infty}^{\infty}\frac{\mathrm{d}k'}{2\pi}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}k'\tilde{\phi}_1(p',k')\tilde{f}_1(\bar{p}',\bar{k}')\right].
\end{aligned} \tag{14.129}$$

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The factors in the convolution integral can be expressed in terms of the original driving potential using Eqs. (14.121) and (14.123) to obtain

$$(p+\mathrm{i}kv)\,\tilde{f}_2(p,k)+\mathrm{i}k\frac{e}{m}\frac{\partial f_0}{\partial v}\tilde{\phi}_2(p,k)=\frac{\partial}{\partial v}\chi(p,k,v), \tag{14.130}$$

where the nonlinear convolution term is

$$\begin{aligned}\chi(p,k,v)=\left(\frac{e}{m}\right)^2\int_{-\infty}^{\infty}\frac{\mathrm{d}k'}{2\pi}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}&\left\{k'\frac{\tilde{\phi}_{ext}(p',k')}{\mathcal{D}(p',k')}\right.\\ &\left.\times\frac{\mathrm{i}\bar{k}'}{\bar{p}'+\mathrm{i}\bar{k}'v}\frac{\tilde{\phi}_{ext}(\bar{p}',\bar{k}')}{\mathcal{D}(\bar{p}',\bar{k}')}\frac{\partial f_0}{\partial v}\right\}.\end{aligned} \tag{14.131}$$

Equation (14.130) may be solved for $\tilde{f}_2(p,k)$ to give

$$\tilde{f}_2(p,k)=-\mathrm{i}\frac{e}{m}\frac{k}{(p+\mathrm{i}kv)}\frac{\partial f_0}{\partial v}\tilde{\phi}_2(p,k)+\frac{1}{(p+\mathrm{i}kv)}\frac{\partial}{\partial v}\chi(p,k,v). \tag{14.132}$$

However, the Fourier–Laplace transform of the second-order Poisson's equation gives

$$-k^2\tilde{\phi}_2(p,k)=\frac{e}{\varepsilon_0}\int\tilde{f}_2(p,k)\mathrm{d}v \tag{14.133}$$

and so substituting for $\tilde{f}_2(p,k)$ gives

$$-k^2\tilde{\phi}_2(p,k)=\frac{e}{\varepsilon_0}\int\left[-\mathrm{i}\frac{e}{m}\frac{k}{(p+\mathrm{i}kv)}\frac{\partial f_0}{\partial v}\tilde{\phi}_2(p,k)+\frac{1}{(p+\mathrm{i}kv)}\frac{\partial\chi}{\partial v}\right]\mathrm{d}v \tag{14.134}$$

or

$$\begin{aligned}\tilde{\phi}_2(p,k)&=-\frac{e}{k^2\varepsilon_0\mathcal{D}(p,k)}\int\frac{1}{(p+\mathrm{i}kv)}\frac{\partial\chi}{\partial v}\mathrm{d}v\\ &=\frac{e}{k^2\varepsilon_0\mathcal{D}(p,k)}\int\frac{\mathrm{i}k\chi}{(p+\mathrm{i}kv)^2}\mathrm{d}v.\end{aligned} \tag{14.135}$$

Substitution for χ and using the velocity-normalized distribution function $\bar{f}_0=f_0/n_0$ discussed in Eq. (14.5) gives

$$\begin{aligned}\tilde{\phi}_2(p,k)=\frac{\omega_p^2}{k^2\mathcal{D}(p,k)}\frac{e}{m}\int\mathrm{d}v\int_{-\infty}^{\infty}\frac{\mathrm{d}k'}{2\pi}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}&\\ \left\{\frac{\mathrm{i}k}{(p+\mathrm{i}kv)^2}\frac{k'\tilde{\phi}_{ext}(p',k')}{\mathcal{D}(p',k')}\frac{\mathrm{i}\bar{k}'}{\bar{p}'+\mathrm{i}\bar{k}'v}\frac{\tilde{\phi}_{ext}(\bar{p}',\bar{k}')}{\mathcal{D}(\bar{p}',\bar{k}')}\frac{\partial\bar{f}_0}{\partial v}\right\}&.\end{aligned} \tag{14.136}$$

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Double-impulse source function and its transform

In order to proceed further, the form of the external source must be specified. We assume the external source consists of two sets of periodic grids, which are pulsed sequentially. The first set of grids has wavenumber k_a and is pulsed at $t=0$ whereas the second set of grids has wavenumber k_b and is pulsed after a delay τ. Thus, the external source has the form

$$\phi_{ext}(x,t) = \phi_a \cos(k_a x)\,\delta(\omega_p t) + \phi_b \cos(k_b x)\,\delta(\omega_p(t-\tau)). \tag{14.137}$$

The Fourier–Laplace transform of this source function gives

$$\begin{aligned}\tilde{\phi}_{ext}(p,k) &= \int_{-\infty}^{\infty}\int_0^{\infty} \phi_{ext}(x,t)\ \mathrm{e}^{-\mathrm{i}kx-pt}\mathrm{d}x\mathrm{d}t \\ &= \frac{\pi}{\omega_p}\sum_{\pm}\{\phi_a\delta(k\pm k_a) + \phi_b\delta(k\pm k_b)\ \mathrm{e}^{-p\tau}\}.\end{aligned} \tag{14.138}$$

Since we are interested in the nonlinear interaction between the a and b pulses, only the contribution from the a pulse in the first $\tilde{\phi}_{ext}$ factor in Eq. (14.136) and the contribution from the b pulse in the second $\tilde{\phi}_{ext}$ factor in Eq. (14.136) will be considered. We therefore substitute the a contribution in Eq. (14.138) for the first $\tilde{\phi}_{ext}$ factor in Eq. (14.136) to obtain

$$\begin{aligned}\tilde{\phi}_2(p,k) = \frac{\omega_p \pi \phi_a}{k^2\mathcal{D}(p,k)}\frac{e}{m}\int \mathrm{d}v \int_{-\infty}^{\infty}\frac{\mathrm{d}k'}{2\pi}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}\frac{\mathrm{i}kk'\delta(k'\pm k_a)}{(p+\mathrm{i}kv)^2\,\mathcal{D}(p',k')} \\ \times\frac{\mathrm{i}\bar{k}'}{(p-p')+\mathrm{i}\bar{k}'v}\frac{\tilde{\phi}_{ext}((p-p'),\bar{k}')}{\mathcal{D}(p-p',\bar{k}')}\frac{\partial \bar{f}_0}{\partial v}.\end{aligned} \tag{14.139}$$

The effect of the $\delta(k'\pm k_a)$ factor when evaluating the k' integral is to force $k' \to \pm k_a$ and also $\bar{k}' \to k \mp k_a$ so

$$\begin{aligned}\tilde{\phi}_2(p,k) = \frac{\omega_p}{k^2\mathcal{D}(p,k)}\frac{e\phi_a}{2m}\sum_{\pm}\int \mathrm{d}v \int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}\frac{(\mp \mathrm{i}kk_a)}{(p+\mathrm{i}kv)^2\,\mathcal{D}(p',\mp k_a)} \\ \times\frac{\mathrm{i}\,(k\mp k_a)}{\bar{p}'+\mathrm{i}\,(k\mp k_a)\,v}\frac{\tilde{\phi}_{ext}(\bar{p}',(k\mp k_a))}{\mathcal{D}(\bar{p}',k\mp k_a)}\frac{\partial \bar{f}_0}{\partial v}.\end{aligned} \tag{14.140}$$

Now let us substitute for the second $\tilde{\phi}_{ext}$ factor using the contribution from the b pulse to obtain

$$\begin{aligned}\tilde{\phi}_2(p,k) = \frac{\pi}{2k^2\mathcal{D}(p,k)}\frac{e}{m}\phi_a\phi_b\sum_{\pm}\sum_{\pm}\int \mathrm{d}v \int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}\frac{(\mp \mathrm{i}kk_a)}{(p+\mathrm{i}kv)^2\,\mathcal{D}(p',\mp k_a)} \\ \times\frac{\mathrm{i}\,(k\mp k_a)}{\bar{p}'+\mathrm{i}\,(k\mp k_a)\,v}\frac{\delta(k\mp k_a\pm k_b)\mathrm{e}^{-\bar{p}'\tau}}{\mathcal{D}(\bar{p}',k\mp k_a)}\frac{\partial \bar{f}_0}{\partial v}.\end{aligned} \tag{14.141}$$

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The upper choice of the $\pm$ and $\mp$ signs is selected and the inverse Fourier transform performed to obtain

$$\begin{aligned}\tilde{\phi}_2^{upper}(p,x) &= \frac{\phi_a\phi_b}{4k^2\mathcal{D}(p,k)}\frac{e}{m}\int_{-\infty}^{\infty}\mathrm{d}k\int\mathrm{d}v\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}\frac{(-\mathrm{i}kk_a)}{(p+\mathrm{i}kv)^2\,\mathcal{D}(p',-k_a)}\\ &\quad\times\frac{\mathrm{i}\,(k-k_a)}{\bar{p}'+\mathrm{i}\,(k-k_a)\,v}\frac{\delta(k-k_a+k_b)\mathrm{e}^{-\bar{p}'\tau+\mathrm{i}kx}}{\mathcal{D}(\bar{p}',k-k_a)}\frac{\partial\bar{f}_0}{\partial v}\\ &= \frac{\phi_a\phi_b}{4\,(k_a-k_b)^2\,\mathcal{D}(p,k_a-k_b)}\frac{e}{m}\int\mathrm{d}v\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}\\ &\quad\times\frac{\mathrm{i}\,(k_a-k_b)}{(p+\mathrm{i}\,(k_a-k_b)\,v)^2}\frac{k_a}{\mathcal{D}(p',-k_a)}\frac{\mathrm{i}k_b}{(\bar{p}'-\mathrm{i}k_bv)}\frac{\mathrm{e}^{-\bar{p}'\tau+\mathrm{i}(k_a-k_b)x}}{\mathcal{D}(\bar{p}',-k_b)}\frac{\partial\bar{f}_0}{\partial v}.\end{aligned}\tag{14.142}$$

The lower choice of the $\pm$ and $\mp$ signs means that $k_a\to -k_a$ and $k_b\to -k_b$ and so at the end of the calculation $\tilde{\phi}_2^{lower}(p,x)$ can also be determined by simply letting $k_a\to -k_a$ and $k_b\to -k_b$. The inverse Laplace transform gives

$$\begin{aligned}\tilde{\phi}_2^{upper}(t,x) &= \frac{\phi_a\phi_b}{4\,(k_a-k_b)^2}\frac{e}{m}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p}{2\pi\mathrm{i}}\frac{1}{\mathcal{D}(p,k_a-k_b)}\int\mathrm{d}v\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\frac{\mathrm{d}p'}{2\pi}\\ &\quad\times\frac{k_a}{(p+\mathrm{i}\,(k_a-k_b)\,v)^2\,\mathcal{D}(p',-k_a)}\frac{\mathrm{i}k_b}{(p-p'-\mathrm{i}k_bv)}\\ &\quad\times\frac{\mathrm{e}^{(pt-(p-p')\tau+\mathrm{i}(k_a-k_b)x)}}{\mathcal{D}(p-p',-k_b)}\frac{\partial\bar{f}_0}{\partial v}.\end{aligned}\tag{14.143}$$

We first consider the p' integral and only retain the ballistic term due to the pole $(p-p')-\mathrm{i}k_bv$. Evaluating the residue associated with this pole gives $p'=p-\mathrm{i}k_bv$ so

$$\begin{aligned}\tilde{\phi}_2^{upper}(t,x) &= -\frac{\phi_a\phi_b}{4\,(k_a-k_b)^2}\frac{e}{m}\int_{b-\mathrm{i}\infty}^{b+\mathrm{i}\infty}\mathrm{d}p\frac{1}{\mathcal{D}(p,k_a-k_b)}\int\mathrm{d}v\\ &\quad\times\frac{\mathrm{i}\,(k_a-k_b)}{(p+\mathrm{i}\,(k_a-k_b)\,v)^2}\frac{\mathrm{i}k_ak_b}{\mathcal{D}(p-\mathrm{i}k_bv,-k_a)}\frac{\mathrm{e}^{pt-\mathrm{i}k_bv\tau+\mathrm{i}(k_a-k_b)x}}{\mathcal{D}(\mathrm{i}k_bv,-k_b)}\frac{\partial\bar{f}_0}{\partial v}.\end{aligned}\tag{14.144}$$

The pole at $(p+\mathrm{i}\,(k_a-k_b)\,v)^2$ is second order, in which case the rule for a second-order residue must be used, i.e.,

$$\oint\mathrm{d}p\frac{1}{(p-p_0)^2}g(p)=\pi\mathrm{i}\lim_{p\to p_0}\frac{\mathrm{d}}{\mathrm{d}p}g(p).\tag{14.145}$$

Thus, we obtain

$$\phi_2(t,x) = -\frac{\phi_a\phi_b\pi\mathrm{i}}{4\,(k_a-k_b)^2}\frac{e}{m}\int \mathrm{d}v \lim_{p\to\mathrm{i}(k_b-k_a)v}$$
$$\frac{\mathrm{d}}{\mathrm{d}p}\left(\frac{\mathrm{i}\,(k_a-k_b)}{\mathcal{D}(p,k_a-k_b)}\frac{\mathrm{i}k_ak_b}{\mathcal{D}(p-\mathrm{i}k_bv,-k_a)}\frac{\mathrm{e}^{pt-\mathrm{i}k_bv\tau+\mathrm{i}(k_a-k_b)x}}{\mathcal{D}(\mathrm{i}k_bv,-k_b)}\frac{\partial\bar{f}_0}{\partial v}\right). \qquad (14.146)$$

The strongest dependence on p is in the exponential e^{pt}, and so retaining only the contribution of this term to the $\mathrm{d}/\mathrm{d}p$ operator gives

$$\begin{aligned}\tilde{\phi}_2^{upper}(t,x) = &-\frac{\pi\mathrm{i}\phi_a\phi_b}{4\,(k_a-k_b)^2}\frac{e}{m}\int \mathrm{d}v \lim_{p\to\mathrm{i}(k_b-k_a)v}\\
&\left(\frac{t\mathrm{i}\,(k_a-k_b)}{\mathcal{D}(p,k_a-k_b)}\frac{\mathrm{i}k_ak_b}{\mathcal{D}(p-\mathrm{i}k_bv,-k_a)}\frac{\mathrm{e}^{pt-\mathrm{i}k_bv\tau+\mathrm{i}(k_a-k_b)x}}{\mathcal{D}(\mathrm{i}k_bv,-k_b)}\frac{\partial\bar{f}_0}{\partial v}\right)\\
= &\frac{\pi}{4}\frac{k_ak_b}{(k_a-k_b)^2}\frac{e\phi_a\phi_b}{m}\mathrm{e}^{\mathrm{i}(k_a-k_b)x}\int \mathrm{d}v\\
&\frac{\mathrm{i}\,(k_a-k_b)\,t}{\mathcal{D}(\mathrm{i}(k_b-k_a)v,k_a-k_b)\,\mathcal{D}(-\mathrm{i}k_av,-k_a)}\\
&\times\frac{\mathrm{e}^{\mathrm{i}(k_b-k_a)vt-\mathrm{i}k_bv\tau}}{\mathcal{D}(\mathrm{i}k_bv,-k_b)}\frac{\partial\bar{f}_0}{\partial v}.\end{aligned} \qquad (14.147)$$

The $\tilde{\phi}_2^{lower}(t,x)$ term is obtained by letting $k_a-k_b\to-(k_a-k_b)$ and so

$$\tilde{\phi}_2(t,x) = \tilde{\phi}_2^{upper}(t,x)+\tilde{\phi}_2^{lower}(t,x). \qquad (14.148)$$

If it is assumed that $\bar{f}_0=(\pi v_T)^{-1/2}\exp\left(-v^2/v_T^2\right)$ then the velocity integrals in the upper and lower terms will be

$$\begin{aligned}\frac{1}{\sqrt{\pi}}\int \mathrm{d}v\mathrm{e}^{-v^2/v_T^2\pm\mathrm{i}[(k_b-k_a)t-k_b\tau]v} &= \exp\left(-\left[(k_b-k_a)t-k_b\tau\right]^2 v_T^2/4\right)\\
&= \exp\left(-\left[t-\frac{k_b}{k_b-k_a}\tau\right]^2\frac{(k_b-k_a)^2v_T^2}{4}\right).\end{aligned} \qquad (14.149)$$

Phase mixing due to the extreme velocity dependence of the $\exp\left(\mathrm{i}(k_b-k_a)vt-\mathrm{i}k_bv\tau\right)$ factor will cause the velocity integral to vanish except at the special time

$$t_{echo} = \frac{k_b}{k_b-k_a}\tau \qquad (14.150)$$

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when there is no phase mixing and so a finite $\tilde{\phi}_2(t,x)$ signal results. This is the echo. The half-width of the echo can be determined by writing Eq. (14.149) as

$$\frac{1}{\sqrt{\pi}}\int \mathrm{d}v \mathrm{e}^{-v^2/v_T^2 \pm \mathrm{i}[(k_b-k_a)t-k_b\tau]v} = \mathrm{e}^{-(t-t_{echo})^2/(\Delta t)^2}, \tag{14.151}$$

where

$$\Delta t = \frac{2}{|k_b-k_a|v_T} \tag{14.152}$$

is the width of the echo.

14.3.5 Spatial echoes

Creating spatially periodic sources with temporal delta functions is experimentally more difficult than creating temporally periodic sources with spatial delta functions. In the latter arrangement, two spatially separated grids are placed in a plasma and each grid is excited at a different frequency. This system is then characterized by a Fourier transform in time and a Laplace transform in space so the convective derivative in the linearized Vlasov equation has the form $-\mathrm{i}\omega f_1 + v\partial f_1 \partial x$ giving ballistic terms proportional to $\exp(\mathrm{i}\omega x/v)$ instead of proportional to $\exp(-\mathrm{i}kvt)$. Thus, if two grids separated by a distance L are excited at respective frequencies ω_1 and ω_2, one grid will excite a ballistic term $\sim \exp(\mathrm{i}\omega_1(x-L)/v)$ while the other will excite a ballistic term $\sim \exp(\mathrm{i}\omega_2 x/v)$, and so the nonlinear product of these two ballistic terms will include a factor $\sim \exp(\mathrm{i}\omega_1(x-L)/v - \mathrm{i}\omega_2 x/v)$, which will phase mix to zero except at the spatial location

$$x_{echo} = \frac{\omega_1 L}{\omega_1-\omega_2}. \tag{14.153}$$

If the spatial Landau damping length of the two linear modes is much less than x_{echo}, then the linear modes will appear to damp away spatially and then the echo will appear at $x = x_{echo}$, which will be much further away. This sort of arrangement was used to demonstrate echoes in lab experiments by Malmberg, Wharton, Gould and O'Neil (1968)

14.3.6 Higher order echoes

If one expands the Vlasov equation to higher than second order, then higher order products of ballistic terms can result. For spatial echoes, high order products of the form

$$\left[\mathrm{e}^{\mathrm{i}\omega_1(x-L)/v}\right]^N \times \left[\mathrm{e}^{-\mathrm{i}\omega_2 x/v}\right]^M \tag{14.154}$$

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will appear and will phase mix away except at locations where

$$N\omega_1(x-L) = M\omega_2 x \tag{14.155}$$

so there will be a higher order echo at the location

$$x_{echo} = \frac{M\omega_2 L}{N\omega_1 - M\omega_2}. \tag{14.156}$$

14.4 Assignments

1. Competition between collisions and quasi-linear diffusion.
   (a) Let $v_{Te} = \sqrt{2\kappa T_e/m_e}$ and $w = v_z/v_{Te}$ and show that the Fokker–Planck equation in Assignment 1 of Chapter 13 can be written in dimensionless form as

$$\frac{\partial F_T}{\partial \tau} \simeq \frac{\partial}{\partial w}\left(F_T \frac{2+Z}{w^2} + \frac{3}{4w^3}\frac{\partial F_T}{\partial w}\right),$$

   where $\tau = \nu t$ and

$$\nu = \frac{n_e e^4 \ln \Lambda}{4\pi\varepsilon_0^2 m_e^2 v_{Te}^3}$$

   is an effective collision frequency.
   Now show that the quasi-linear diffusion equation can similarly be written in dimensionless form as

$$\frac{\partial F_T}{\partial \tau} = \frac{\partial}{\partial w}\left(\bar{D}_{QL}\frac{\partial F_T}{\partial w}\right),$$

   where

$$\bar{D}_{QL}(w) = \frac{D_{QL}}{D_{\mathrm{col}}}$$

   and

$$D_{\mathrm{col}} = \frac{2\nu\kappa T_e}{m_e}$$

   is an effective collisional velocity-space diffusion coefficient.
   (b) Suppose both collisions and quasi-linear diffusion are operative so

$$\frac{\partial F_T}{\partial \tau} = \frac{\partial}{\partial w}\left(\bar{D}_{QL}\frac{\partial F_T}{\partial w} + F_T\frac{2+Z}{w^2} + \frac{3}{4w^3}\frac{\partial F_T}{\partial w}\right).$$

   Show that the steady-state solution to this equation has the form (Fisch 1978)

$$F_T \sim \exp\left(-\int^w \mathrm{d}w \frac{(2+Z)\,w}{\left(w^3\bar{D}_{QL} + \frac{3}{4}\right)}\right).$$

   Sketch F_T for the cases where (i) $\bar{D}_{QL} \gg 1$ over a certain range of velocities (i.e., where wave spectrum is resonant with particles) and (ii) $\bar{D}_{QL} \ll 1$. How could this

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be used to drive a toroidal current in a tokamak? Using the appropriate form of D_{QL} show that the normalized quasi-linear diffusion for resonant particles is

$$\bar{D}_{QL}(w) = \frac{8\pi^2 \mathcal{E}(\omega_r/v,t)}{w m_e \omega_{p_e}^2 \ln\Lambda}.$$

If the wave is electrostatic then show

$$\mathcal{E}(\omega_r/v,t) \sim k|\phi|^2,$$

where k is the wavenumber of the waves resonantly interacting with the particles so

$$\bar{D}_{QL}(w) \sim \frac{8\pi^2 k|\phi|^2}{m_e w \omega_{p_e}^2 \ln\Lambda}.$$

What sort of waves would be most efficient at driving current, large or small k? Would this current drive work best in high- or low-density plasmas?

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